SWEEP rule on the KD-Toric code

April 15, 2026

1 Shrinking

consider changing the name to something like hight lines / contour in order to hint on the similarity to topography map. In addition, the embedding inside $\mathbb{R}^d$ should appears before that definition.

Definition 1.1 (Potential Walls)**.** We define the potential walls to be a functions collection from the vertices of $\mathcal{G}_n^d$ to the reals as follow:

1. For any coordinate $i \in [d]$, we define $L_i \colon \mathcal{G}_n^d \to \mathbb{R}$ to act as $L_i(v) = v_i$.
2. For any subset of coordinates $I \subset [d]$ we define $L_I \colon \mathcal{G}_n^d \to \mathbb{R}$ to act as $L_I(v) = -\sum_{i \in I} v_i$.

For a subset of vertices $\Gamma \subset \mathcal{G}_n^d$, a vertex $v \in \Gamma$, and a potential wall $l \colon \mathcal{G}_n^d \to \mathbb{R}$, we say that v is a *local maximum* of l in Γ if for any neighbor u of v, such that $u \in \Gamma$, we have $l(v) \geq l(u)$. We say that v is a strong local maximum if the inequality is strict, namely $l(v) > l(u)$. When the subset Γ is clear from the context, we might omit it and briefly say that v is a local maximum of l.

What we want is that if a bit contains both v, gv then there will be a check that contains both v, gv, that is true if $d' > 1$, then it means that we can earse from the bit a vertex which it is not gv.

Claim 1.2. Let z be an assignment, and let ∂z be the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂z, and let g be a coordinate in S such that v is a local maximum of L_g. Then $gv \notin \partial z$.

Proof. Suppose $gv \in \partial z$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in ∂z. $\square$

Claim 1.3. Let z be an assignment, and let ∂z be the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂z, and let g be a coordinate in S such that v is a local maximum of L_g. Then for any check associated with a $d' - 1$ face rooted at v that contains gv, it is satisfied.

Proof. Assume, through contradiction, the existence of an unsatisfied check associated with a $d' - 1$ positive face rooted at v, which contains gv. This implies $gv \in \partial z$, contradicting Claim 1.2. □

Definition 1.4 (Small Code at Vertex v)**.** For a vertex v, we define the *X-small code at vertex v* to be the binary assignments over the bits (d' faces) rooted at v that satisfy $\mathcal{C}_X$'s restrictions rooted at v ($d' - 1$ faces). Similarly, we define the *Z-small code at vertex v* to be the binary assignments over the bits (d' faces) rooted at v that satisfy $\mathcal{C}_Z$'s restrictions rooted at v ($d' + 1$ faces). We denote by $\mathcal{C}_v^-$ and $\mathcal{C}_v^+$ the X-small code at v and Z-small code at v.

Definition 1.5. Should rewrite it better. We define Stab_v^- and Stab_v^+ as the following linear codes:

$$\begin{aligned}\mathrm{Stab}_v^- := \big\{z : \partial z = 0 \text{ and} \\ z \in \mathrm{Span}\left(\left\{\theta_{\mathbf{v},\mathbf{S'/g}}, \theta_{\mathbf{gv},\mathbf{S'/g}} \text{ for } S' \subset S, |S'| = d' + 1, g \in S'\right\}\right)\big\}\end{aligned}$$

$$\begin{aligned}\mathrm{Stab}_v^+ := \big\{z : \partial^* z = 0 \text{ and} \\ z \in \mathrm{Span}\left(\left\{\theta_{\mathbf{v},\mathbf{S'/g}}, \theta_{\mathbf{gv},\mathbf{S'/g}} \text{ for } S' \subset S, |S'| = d' + 1, g \in S'\right\}\right)\big\}\end{aligned}$$

Claim 1.6. Let v be a vertex. For any $z \in \mathcal{C}_v^\pm$, there is a stabilizer $w \in \mathrm{Stab}_v^\pm$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v.

Proof. We will show that codes obtained from $\mathrm{Stab}_v^\pm$ by forcing the faces rooted at v to $z|_v$ have non-zero dimensions. Then it follows that any code word of $\mathcal{C}_v^\pm$ could be extend to codewords in $\mathrm{Stab}_v^\pm$. The number of bits in both $\mathrm{Stab}_v^\pm$ is:

$$\binom{d}{d'} + d\binom{d-1}{d'}$$

The numbers of checks for $\mathrm{Stab}_\pm$ are:

$$\binom{d}{d' \pm 1} + d\binom{d-1}{d' \pm 1}$$

The restriction to $z|_v$ sets the bits rooted at v and satisfies the checks that are rooted at v, so the checks of the form $\theta_{\mathbf{v},\mathbf{S'}}$ for $|S'| = d' \pm 1$ are degenerate, and the dimension that remains is:

$$\dim \geq d\binom{d-1}{d'} - d\binom{d-1}{d' \pm 1}$$

In the case that $d' = (d-1)/2$, then for any $x \in [d-1]$ we have $\binom{d-1}{d'} > \binom{d-1}{x}$ therefore:

$$\begin{aligned}\dim &\geq d\binom{d-1}{d/2} - d\binom{d-1}{d/2 \pm 1}\\ &> d\end{aligned}$$

So there is at least 2^d stabilizers $w \in \mathrm{Stab}_v$ that agree with z. $\square$

Claim 1.7. Let z be a finite assignment. Suppose that

$$\max\left\{L_{d+1}(u) : u \in z\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1$$

Proof. Let $v_1, v_2, \dots, v_k$ be the vertices in z that achieve the maximum for L_{d+1}, namely $v_1, v_2, \dots, v_k = \arg\max_{v \in z} L_{d+1}(v)$, and assume that $v_i \notin \partial^{\pm} z$ for any $i \in [k]$.

Consider the ith vertex v_i. By definition, since $v_i \notin \partial^{\pm} z$, we have $\partial^{\pm} z|_{v_i} = 0$, and since v_i is a maximum point of L_{d+1} in z, it is also a local maximum, and therefore $z|_{v_i+} = z|_{v_i}$. Combining the two, we get that $z|_{v_i}$ is a codeword of the small code at v_i, namely $z|_{v_i} \in \mathcal{C}^{\pm}_{v_i}$. Thus, by Claim 1.6, there is a stabilizer $w_i \in \mathrm{Stab}^{\pm}_{v_i}$ such that w_i and z agree on the positive faces of v_i, namely $w_i|_{v_i} = z|_{v_i}$.

Set $\tilde{z} \leftarrow z + \sum_i w_i$. Since $\sum_i w_i$ is a stabilizer, we have $\tilde{z} \sim_{C_X} z$. In addition, for any $j \neq i$, it follows that $v_j \notin w_i$. Otherwise, there exists $S' \subset S$ such that $v_j \in \theta_{v_i, S'}$, and therefore $L_{d+1}(v_j) > L_{d+1}(v_i)$, in contradiction to v_i and v_j being both maximum points for L_{d+1} in $\partial^{\pm} z$. Thus, we have:

$$\tilde{z}|_{v_i} = (z + \sum_i w_i)|_{v_i} + (z + w_i)|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{z}$. Hence, the vertex domain of $\tilde{z}$ is contained in the union of the vertices of z with the vertices of $w_1, w_2, \dots, w_k$, substituting $v_1, v_2, \dots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} &\leq \max\left\{L_{d+1}(u) : u \in z \bigcup_i w_i / \bigcup_i v_i\right\}\\ &\leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1\end{aligned}$$

$\square$

Claim 1.8. Let z be a finite assignment. Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Proof. Let z be a finite assignment, and let k be an integer such that $k \geq 1$ and

$$\max\left\{L_{d+1}(u) : u \in z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} + k$$

For an integer $l \leq k$, consider the assignments $\tilde{z}_0, \tilde{z}_1, .., \tilde{z}_l$ defined as follow:

- The first element, $\tilde{z}_0 = z$.
- If the $i-1$th element satisfies

$$(\star) \quad \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}_{i-1}\right\}$$

  Then define the ith element, $\tilde{z}_i$, to be the result of applying the construction, given in Claim 1.7, on $\tilde{z}_{i-1}$, namely $\partial\tilde{z}_{i+1} = \partial z_i$ and

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}_i\right\} \leq \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} - 1$$

  If the $i-1$th element doesn't satisfies $\star$, then set $l = i-1$.
- If the kth element was defined, set $l = k$.

Since $\sim_{C_X}$ is a transitive relation, we have that for any $i \in [l]$, $z \sim_{C_X} \tilde{z}_i$. If $l \neq k$, then there exists $\tilde{z}_i$ such that $\tilde{z}_i \sim_{C_X}$ and $\tilde{z}_i$ satisfies $\star$, namely by setting $\tilde{z} = \tilde{z}_i$, we get the desired. So, it is left to handle the case where $l = k$. In that case, set $\tilde{z} = \tilde{z}_k$. By Claim 1.7:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Yet, since $\tilde{z} \sim_{C_X} z$, it follows that $\partial\tilde{z} = \partial z$, So:

$$\max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\}$$

Therefore, we have the equality:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

□

Claim 1.9. Let z be a finite assignment and let ξ be the result of the application of the Sweep rule on it. Then:

1. For any $i \in [d]$: $\max\left\{L_i(v) : v \in \partial z\right\} = \max\left\{L_i(v) : v \in \partial\xi\right\}$
2. And for $i = d+1$: $\max\left\{L_{d+1}(v) : v \in \partial z\right\} > \max\left\{L_{d+1}(v) : v \in \partial\xi\right\} + 1$

Proof. First, since the Sweep rule is invariant to addition by stabilizers, for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, the correction applied by the Sweep rule on $\tilde{z}$ is the same as the application on z. Denote by ϕ the correction, by $\tilde{\xi}$ the result of applying the Sweep rule on $\tilde{z}$, and by w the stabilizer which is the difference between z and $\tilde{z}$, namely:

$$\xi = z + \phi$$
$$\tilde{\xi} = \tilde{z} + \phi$$
$$\tilde{z} = z + w$$

The $\partial^{\pm}$ boundary of $\tilde{\xi}$ equals:

$$\partial\tilde{\xi} = \partial\left(\tilde{z} + \phi\right) = \partial\left(z + \phi\right) = \partial\xi$$

So, in total, we have that for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, it holds that $\partial\tilde{z} = \partial z$ and $\partial\tilde{\xi} = \partial\xi$. Therefore, showing that there exists $\tilde{z} \sim_{C_X} z$ for which the claim holds proves the claim for z. By Claim 1.8, there is $\tilde{z} \sim_{C_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}\right\}$$

For each $\tilde{z}$, any maximum of L_{d+1} in $\partial\tilde{z}$, is also a maximum in $\tilde{z}$. Denote by $v_1, .., v_k$ the maximum points in $\partial\tilde{z}$. Since they are maximum points in $\tilde{z}$, it holds that for each $i \in [k]$, $\tilde{z}|_{v_i} = \tilde{z}|_{v_i+}$, and hence $\partial\tilde{z}|_{v_i} = \partial\tilde{z}|_{v_i+}$. Since $\tilde{z}|_{v_i+}$ is an assignment rooted at v_i, then v_i can suggest $\tilde{z}|_{v_i+}$, namely the condtion in Sweep rule is satisfied for vertex v_i. Thus:

$$\partial\tilde{\xi}|_{v_i} = \partial\tilde{z}|_{v_i} + \partial\phi|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{\xi}$. Hence, the vertex domain of $\tilde{\xi}$ is contained in the union of the vertices of $\tilde{z}$ with the vertices of $\phi_1, \phi_2, \ldots, \phi_k$, substituting $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} &\leq \max\left\{L_{d+1}(u) : u \in z\bigcup_i w_i / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1\end{aligned}$$

□

Proof.

$$L_{d+1}(u) \leq \min_{v \in \partial z} L_{d+1}(v)$$

Observes that if the *termination-condition* holds then any local maximum of L_{d+1} in ∂z is also a local maximum of L_{d+1} in $\tilde{z}$ and we finish. Consider the other case, namely there is at least one vertex $u \in \tilde{z}/\partial z$ such that u is a local maximum for L_{d+1} and

$$L_{d+1}(u) > \min_{v \in \partial z} L_{d+1}(v)$$

In that case, we peak such u, and looking for stabilizer w such $u \notin \tilde{z} + w$ by repeating over the above procedure where u plays the role of v. We keep repeating on the procedure until the *termination-condition* holds.

We argue that the process indeed terminates. Consider a vertex u such that u is a local maximum of L_{d+1} in $\tilde{z}$ and u is not a local maximum of L_{d+1} at z. Put differently, u becomes a local maximum as a result of adding w to z. Thus, u has to belong to $w|_v \setminus v$, which implies that there is a non-empty subset of generators $I \subset S$ such that $u = (\prod_{g \in I} g)v$, and therefore $L_{d+1}(u) \leq -1 + L_{d+1}(v)$.

Hence, since z is finite.

We repeat the process while possible; namely, we stop when all the local maxima of L_{d+1} are in ∂. □

We prove that $\dim \mathcal{C}_v = \dim \mathrm{Stab}_v$ below, for now assume it is correct. We continue by using the fact that if, for a local maximum v in ∂, it holds that $z|_v$ is rooted at v, then the decoder will find bits rooted at v such that their ∂ agrees with ∂.

To generalize, one might think that it has to complete the local view on v such that all the checks are satisfied. For C_X, it sums up to show that the following dimension is positive. Yet the trick fails for C_Z, where we need to think of a different idea.

$$d\binom{d-1}{d'} - \binom{d}{d'-1} - d\binom{d-1}{d'-1}$$

Claim 1.10. Let z an assignment rooted at vertex v, such $\partial z|_{v+} = 0$. Then there is a stabilizer w, rooted at v, such that $w|_{v+} = z$.

Proof. Denote by F the faces in z, and by J the collection of generators that support F. We claim that $|J| \geq d' + 1$.

Clearly, $|J| \geq d'$, since each face is defined by a root vertex and d' generators. Yet, observe that if $|J|$ equals exactly d', then z contains a single d'-face, denote it by f. Then, any check associated with a $d' - 1$ face of f is unsatisfied, in particular the $d' - 1$ faces that are rooted at v, in contradiction to $\partial z|_{v+} = 0$.

Now, one can choose $J' \subset J$ of size $d' + 1$, and define the stabilizer $w = \theta_{v,J'}$. □

Claim 1.11. Suppose $v \in z$ and for any $g \in S$, we have $g^{-1}v \notin z$, then there is $\theta_{v,J} \in \partial z$.

Proof.

$$\partial z = \sum_{S'} \theta_{\mathbf{v},\mathbf{S'}} + ..$$

$\square$

Claim 1.12. Let z be a connected assignment on the d'-faces at finite size, and let Γ be the set of vertices which are adjacent to unsatisfied checks induced by z. Let v be a vertex in $\mathcal{G}_\infty^d$ such that v is a strong global maximum of L_{d+1}, at Γ. Then there is a rooted assignment $\tilde{z}$ at v such that $z_v = \tilde{z}_v$.

Proof. Let $\Theta = \theta_{v_1,S'_1}, \theta_{v_2,S'_2}, ..\theta_{v_k,S'_k}$ be the faces forms z.

Since z is finite, it has a maximum for L_{d+1}. Denote by u the vertex that gets the maximum, and denote by $\theta_{u,J_1}, \ldots, \theta_{u,J_l}$ the d'-faces in z that are rooted at u. We claim that $u = v$.

Suppose not, namely $u \notin \Gamma$. Then it means that any $d' - 1$ face containing u is satisfied. Observe that $\sum \theta_{u,J_i} \neq 0$; otherwise, it would contradict the non linearity independence of $\theta_{\mathbf{v_1},\mathbf{S'_1}}, \theta_{\mathbf{v_2},\mathbf{S'_2}}, ..\theta_{\mathbf{v_k},\mathbf{S'_k}}$. So, by Claim 1.11 there exists a $d' - 1$ face rooted at u, contained in $\sum \theta_{u,J_i}$. Let us denote it by $\theta_{u,J'}$.

Since $\theta_{u,J'}$ is satisfied, it has to be adjacent to a turned bit, namely a d' face in z, not in $\theta_{u,J_1}, \ldots, \theta_{u,J_l}$. Denote that d' face by f. Yet, since $\theta_{u,J'}$ is rooted at u, it contains u, and therefore f contains u. Now, if f is rooted at u, then it is a contradiction that $\theta_{u,J_1}, \ldots, \theta_{u,J_l}$ are all the d' faces in z rooted at u. If f is not rooted at u, denote its root by w, so there is $g \in S$ such that $gw = u$, and $w \in z$, in contradiction to u being maximum for L_{d+1}.

So, we find that v is also the maximum of L_{d+1} in z. Therefore, we have that $\tilde{z} = \sum \theta_{u,J_i}$ zeros out z_v. $\square$

Claim 1.13. Let z be a connected assignment on the d'-faces at finite size, and let the collection $\Theta = \theta_{v_1,S'_1}, \theta_{v_2,S'_2}, ..\theta_{v_k,S'_k}$ be the decomposition forms z. Denote by Γ the set of vertices adjacent to unsatisfied check induced by z. Then Θ is acyclic, if and only if z has at least one Infimum.

Proof. First, suppose that z has at least one Infimum. Let u be an infimum in z.. Second, suppose that Θ is acyclic. □

Claim 1.14. Suppose that $d' > 1$ and a vertex $v \in \mathcal{G}^d_\infty$ is also adjacent to the domain wall, namely $v \in \Gamma$. It holds that v cannot be simultaneously both a (strong) local max of L_{d+1} and a local min of L_{d+1}.

Proof. Let $v \in \Gamma$ be a local maximum point of L_{d+1}, It means that:

1. The vertex v is adjacent to a d'^{-1}-face which is unsatisfied.
2. That face is rooted in v; otherwise, the root of the face is a vertex in Γ with a higher value of L_{d+1}, in contradiction to v being a local maximum.

Thus, there is a subset of generators $S' \subset S$, of size $d' - 1 > 0$, such that the unsatisfied check set is on the face $\theta_{v,S'}$. Given $d' > 1$, we have that S' contains at least one more element besides the empty set, denoted by g. Now, consider the vertex gv. Since gv is adjacent to $\theta_{v,S'}$, then $gv \in \Gamma$. On the other hand, on the infinite toric $\mathcal{G}^d_\infty$, moving along the grid generator decreases L_{d+1} by one, so $L_{d+1}(gv) = L_{d+1}(v) - 1$. In other words, gv is a neighbor of v with a lower value of L_{d+1}, and thus v cannot be a local minimum of L_{d+1}. □

Remark 1.15. That claim alone gives a decoder in the clean setting. Write a remark indicating that when $d' = 1$, meaning qubits are on the edges (2D toric), the claim is not true.

For $d = 4, d' = 2$. For $i, j \in [d]$, such that $i \neq j$, denote the bit rooted at v and spanned by generators $g_i, g_j \in S$, namely $\theta_{v,\{g_i,g_j\}}$, by $g_{ij} = g_{ji}$. The restrictions:

$$r_i \colon \sum_{j \neq i} g_{ij}$$

Thus we get

$$\begin{aligned}\sum_i^n r_i &= \sum_i^n \sum_{j\neq i}^{n+1} g_{ij} \\ &= \sum_i^n \sum_{j<i}^{n} g_{i,j} + g_{j,i} + \sum_i^n g_{i,n+1} \\ &= r_{n+1}\end{aligned}$$

Thus the dimension of the small code at v is at least:

$$\binom{d}{2} - (d-1) = 3$$

We have $\binom{4}{3} = 4$ stabilizers rooted at v (each for each cube). Yet:

$$
\begin{aligned}
s_1 &= g_{1,2} + g_{2,3} + g_{1,3} \\
s_2 &= g_{1,2} + g_{2,4} + g_{1,4} \\
s_3 &= g_{1,3} + g_{3,4} + g_{1,4} \\
s_4 &= g_{2,3} + g_{3,4} + g_{2,4}
\end{aligned}
$$

And clearly

$$s_4 = s_1 + s_2 + s_3$$

Definition 1.16 (Reduced subset with boundrey at v)**.**